# Chapter 1

# Natural Community Types: Partitioning Natural Variation and Associating with Environmental Descriptors

## 1.1 Introduction

Before the contamination stress influence on the community can be examined in isolation, the influence of the natural environment must be taken into account. The community records of interest at any point in space and time are the joint outcome of natural environmental conditions, contamination pressure, and other, unrecorded sources of stress(Allan, 2004 [1]). Therefore a site's taxa record is a representation of all these factors acting together.

Two strategies are commonly used: adjusting for the natural environment within the model, or stratifying the data by it before the model is fitted. The first introduces the environmental variables as covariates in the regression of the community records of interest, so that contamination and natural conditions are modelled jointly and estimated finely; its drawback is that contamination gradients typically covary with natural environmental gradients (Graham, 2003 [3]), resulting collinearity with unstable coefficients and limited practical value. The second strategy instead treats the natural environment as a categorical descriptor and partitions the sites into classes of similar environmental and(or) regional—characteristics (Hughes, Larsen, and Omernik, 1986 [4]; Allan, 2004 [1]) to avoid introducing covariates. This stratification can be a more interpretable approach, since it isolates the contamination signal within definable and comparable habitat stratas and divides the community records of interest with natural variation into community types with aligned environmental conditions.

However, the difficulty is practical rather than conceptual. Natural community composition is shaped by a large set of environmental features, many of which are unobservable, observable but not recorded, or recorded only with substantial measurement uncertainty. Consequently, in most empirical settings, the limited set of measured environmental variables is insufficient to define site groups that are genuinely homogeneous in both environmental conditions and the community records of interest. This limitation is illustrated in Figure 1.1. Complete environmental observation aligns the partition in $\Omega_{env}$ with the true community boundary; incomplete observation yields only a degraded partition that mislabels the sites between it and the true boundary—an error of incomplete descriptors, not of a failed environment–community relationship.

For this reason, we reverse the order in which environment and community enter the stratification. We first let the community data *define* the types directly, and then ask what environmental signature *describes* them. It rests on the previous conclusion: within the least-contaminated set, the recorded assemblages retain natural community composition. Therefore we can read natural community types from these least-contaminated sites, since they carry natural variation that we wish to stratify by. Starting with the least-contaminated set, we cluster its community matrix, validate and retain $K$ branches as natural community types. Next, we model this environment-assemblage association by training a classifier that maps their environmental descriptors onto the defined community types. Through this process, we uncanonically stratify community types on the taxa space $\Omega_{taxa}$, makes the implicit environment-determination assumption become an explicit and measurable property in the classifier building. This *define*-then-*describe* workflow is illustrated in Fig-

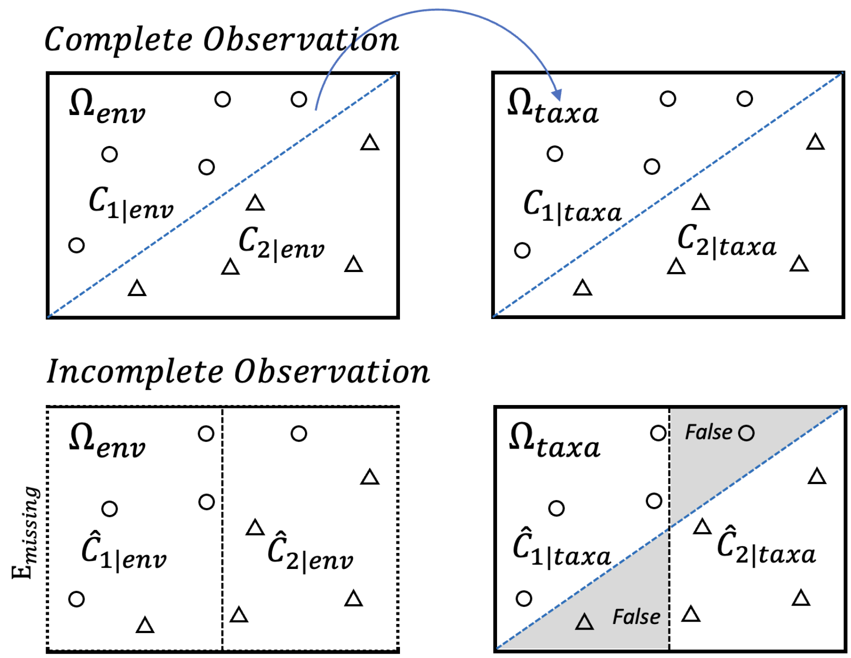


**Figure 1.1:** Under the assumption that environment determines community, incomplete environmental observation propagates classification error from $\Omega_{env}$ to $\Omega_{taxa}$. Complete observation recovers the true diagonal boundary; incomplete observation imposes a degraded vertical partition, mislabelling sites in the wedge between the two (shaded, "False").

ure 1.2.

The built classifier is then applied to the remaining, contaminated sites: predicting their environmental descriptors into community types. Through this process, every site regardless of contamination status is assigned to its natural community type in a broad sense.

## 1.2 Methods

The overall aim of this stage is to assign every site, regardless of contamination level, to a natural community type. The procedure has three main steps, the first two carried out entirely within the least-contaminated set and the third applying their results to the remaining sites. First, natural community types are defined on the least-contaminated sites by clustering their community matrix. Second, starting from these

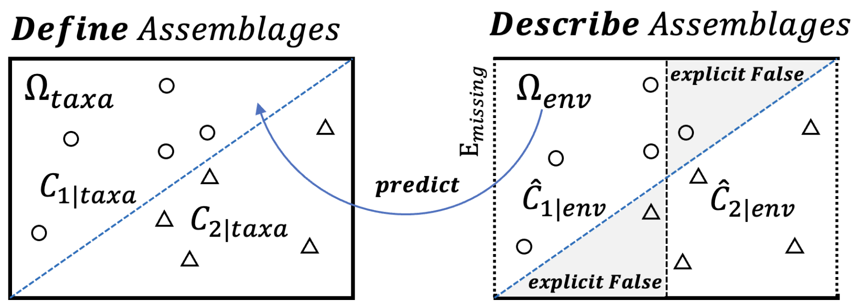


**Figure 1.2:** Community types are *defined* on the taxa space $\Omega_{taxa}$, then *described* and predicted from the environmental space $\Omega_{env}$. Limited environmental features make some misassignments, but these are rendered explicit (shaded, "explicit False").

defined types, a discriminating subset of environmental descriptors is identified and a classifier is built to quantify the relationship between environmental features and community types. Third, the classifier is applied to the remaining contaminated sites, predicting and delineating the decision boundary by which they are assigned to the corresponding natural community type. The remainder of this section describes each step in detail.

### 1.2.1 Unconstrained community typing and its statistical validation

#### Dissimilarity between sites, Double-zero problem and Hellinger transformation

Hierarchical clustering operates on a matrix of pairwise dissimilarities, so the first choice is how dissimilarity between sites is measured from the community records. Applying the Euclidean distance directly to raw abundances is inappropriate for community data, because it treats the joint absence of a taxon at two sites as evidence of similarity—the *double-zero problem*—and lets a few abundant taxa dominate the distance. To avoid this, the community matrix is first Hellinger-transformed: at a given site, each taxon's record is divided by the site's total and the square root is taken, so that the entry becomes the square root of the taxon's relative abundance at that site. The Euclidean distance computed between two Hellinger-transformed site profiles is then equal to the Hellinger distance between their original compositions, a measure that is insensitive to double zeros and to differences in total abundance, and which is recommended for ordination and clustering of community data in numerical ecology (Borcard, Gillet, and Legendre, 2018 [2]). This transform therefore makes the subsequent Euclidean-based clustering ecologically meaningful.

### Hierarchical clustering with Ward's method

The Hellinger-based dissimilarities are clustered by agglomerative hierarchical clustering. Among the available linkage functions, Ward's method is chosen because it merges, at each step, the pair of clusters whose fusion produces the smallest increase in within-cluster variance of the Euclidean (here Hellinger) distances. This criterion suits the goal of delineating community types with members mutually similar in composition. The procedure yields a dendrogram in which the height of each fusion records the linkage function value across which the corresponding branches join; cutting the dendrogram at a chosen linkage height then defines the $K$ community types retained for the subsequent analysis.

### Statistical evidence for the branching: node-wise and overall ANOVA

A dendrogram will partition any data, so the branching structure requires statistical support. Two complementary tests are used, both applied to the Hellinger-transformed taxa abundances used in the clustering rather than to the raw counts. First, a *node-wise* ANOVA is applied from the first to $K + 2$ branching nodes of the dendrogram: for every taxon, its Hellinger-transformed abundances in the two daughter branches are compared, identifying the taxa whose mean values differ significantly across the split. These taxa are the indicator species that contribute most to the formation of that particular node and drive the bifurcation. Second, once the $K$ community types are finalised, an overall ANOVA tests, for each taxon, whether its Hellinger-transformed mean abundance differs significantly among the $K$ groups, providing statistical evidence that the retained partition separates the community types along interpretable compositional gradients.

### Robustness of the partition with pvclust

Beyond the compositional evidence for each branch, the stability of the branches under resampling is assessed with pvclust (Suzuki and Shimodaira, 2006 [5]; Suzuki, Terada, and Shimodaira, 2019 [6]). The method repeatedly perturbs the descriptor space by multiscale bootstrap resampling of the taxa, re-runs the same clustering procedure on each resampled space, and checks whether each branch of the original (anchor) dendrogram reappears. From the frequency with which a branch recurs across the $B$ bootstrap replicates, an *Approximately Unbiased* (AU) $p$-value is computed for that branch. A high AU $p$-value means the branch is recovered in a large proportion of the perturbed spaces—its existence does not depend on a few particular taxa but is supported by many. The ordinary *Bootstrap Probability* (BP), estimated from fixed-size bootstrap resampling alone, is also reported, but it is a more biased than the AU value. Branches with low AU support are treated as uncertain and inform the choice of where the dendrogram is cut. Figure 1.3 illustrates the computation workflow [1], more details about this procedure can be found in the articles by Suzuki and Shimodaira (Suzuki and Shimodaira, 2006 [5]).

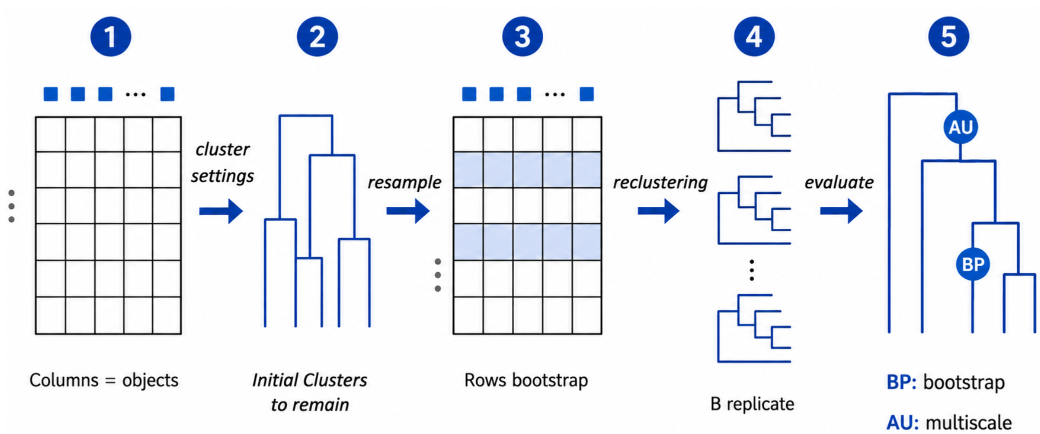


**Figure 1.3:** The pvclust workflow: bootstrap-resample taxa, recluster $B$ times, and score each anchor branch by its AU (multiscale) and BP (bootstrap) recurrence.

[1]The data matrix is transposed so that taxa occupy the rows, because resampling routines in R and similar environments operate row-wise; bootstrapping the feature variables therefore requires placing them as rows.

### 1.2.2 Linking community types to environmental descriptors: classifier training under limited features

#### Limited environmental features and the 2008 reference-typing solution

A key difficulty of the present design persists even after the community types are defined: the limited set of measurable environmental features. Types delineated by unconstrained clustering may remain entangled in environmental space, where a missing key variable, or attention to uninformative ones, blurs the environment–type relationship. This already surfaced in the 2008 corridor study.

There, the Ward-clustered community types could not be cleanly separated on the five available environmental variables, so a directly trained classifier would have performed poorly. The solution was to define the two types by a selective assignment of branches. As shown in Figure 1.7, the dendrogram first divides into $B_2$ versus the pair $\{B_1, A\}$, and only at a lower height do $B_1$ and $A$ separate. Cluster $C_1$ was assigned to $B_2$ and $B_1$ jointly, and Cluster $C_2$ to $A$ alone—a definition under which the types separate more readily on the available variables.

The cost is that the types no longer reflect the dendrogram. The assignment is *non-monophyletic*: $C_1$ joins $B_2$ and $B_1$, two branches that do not form a single subtree, while the topologically closer $A$ is placed in $C_2$. No single horizontal cut can produce this grouping; the labels are assigned across the tree's natural splitting order. The 2008 typing thus lets environmental separability override the community structure the clustering reveals, and the classifier's resulting accuracy is correspondingly inflated (Figure 1.8).

In the descriptor-space view, this operation amounts to relabelling the clustering-defined sites: circles ($C_1$) that sit, on the available features, among the triangles ($C_2$) are reassigned to $C_2$, buying environmental separability at the cost of the true taxa labels (Figure 1.4).

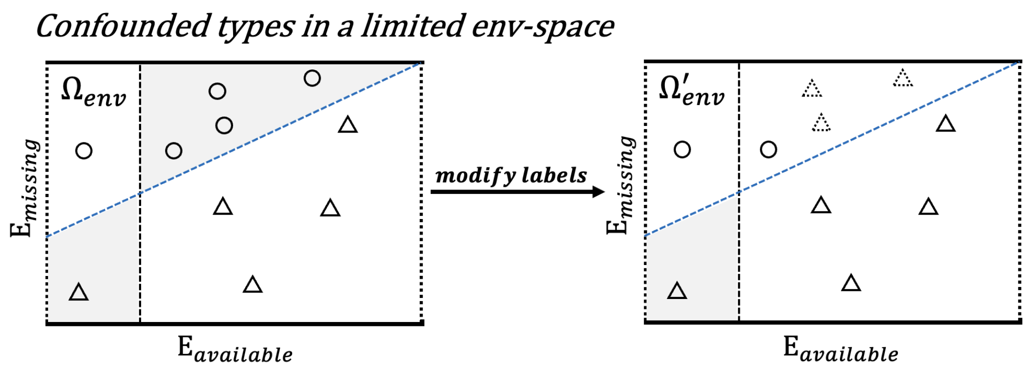


**Figure 1.4:** Confounded community types in a limited environmental space, illustrating the 2008 approach of Zhang. The true type boundary (blue dashed) is governed by an unobserved variable $E_{missing}$—its slope, rather than a diagonal, signifies that the missing variable is the more decisive one—so on the available features $E_{available}$ the types overlap (shaded). Zhang's solution relabels part of $C_1$ (circles) as $C_2$ (triangles) in one direction only ($\Omega_{env} \to \Omega'_{env}$, dotted triangles), reducing but not removing the overlap—separability bought by altering the true cluster labels.

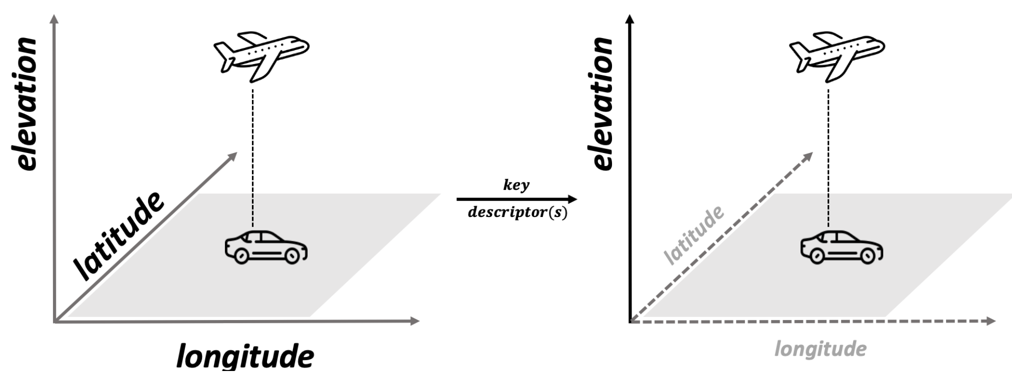


**Figure 1.5:** Not all descriptors help separate a small number of classes. An aircraft and a car are barely distinguishable on the horizontal descriptors (longitude, latitude), yet a single key descriptor—elevation—separates them cleanly. When only a few classes must be told apart, the right descriptor subset matters more than the number of descriptors, and retaining irrelevant ones only adds noise.

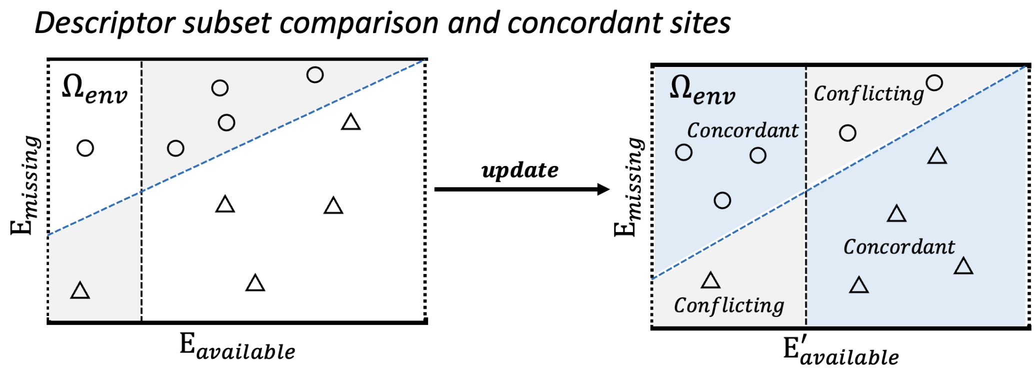


**Figure 1.6:** Selecting a descriptor subset and identifying concordant sites. On the initial descriptors $E_{available}$, the taxa-defined types overlap (left). Choosing a more discriminating subset $E'_{available}$ (right) separates most sites into *concordant* regions (blue), where taxa membership and environmental position agree, while the residual *conflicting* regions (grey) mark sites whose two views disagree. Only concordant sites are used to train the classifier; conflicting sites are retained as diagnostics. In practice, concordance is resolved across four categories (concordant, taxa-only, env-only, ambiguous), here simplified to two.

# References


[1] J. David Allan. “Landscapes and Riverscapes: The Influence of Land Use on Stream Ecosystems”. In: *Annual Review of Ecology, Evolution, and Systematics* 35 (2004), pp. 257–284. DOI: `10.1146/annurev.ecolsys.35.120202.110122`.

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[3] Michael H. Graham. “Confronting Multicollinearity in Ecological Multiple Regression”. In: *Ecology* 84.11 (2003), pp. 2809–2815. DOI: `10.1890/02-3114`.

[4] Robert M. Hughes, David P. Larsen, and James M. Omernik. “Regional Reference Sites: A Method for Assessing Stream Potentials”. In: *Environmental Management* 10.5 (1986), pp. 629–635. DOI: `10.1007/BF01866767`.

[5] Ryota Suzuki and Hidetoshi Shimodaira. “Pvclust: An R Package for Assessing the Uncertainty in Hierarchical Clustering”. In: *Bioinformatics* 22.12 (2006), pp. 1540–1542. DOI: `10.1093/bioinformatics/btl117`.

[6] Ryota Suzuki, Yoshikazu Terada, and Hidetoshi Shimodaira. *pvclust: Hierarchical Clustering with P-Values via Multiscale Bootstrap Resampling*. R package version 2.2-0. 2019. URL: `https://CRAN.R-project.org/package=pvclust`.


### 1.2.3 Appendix

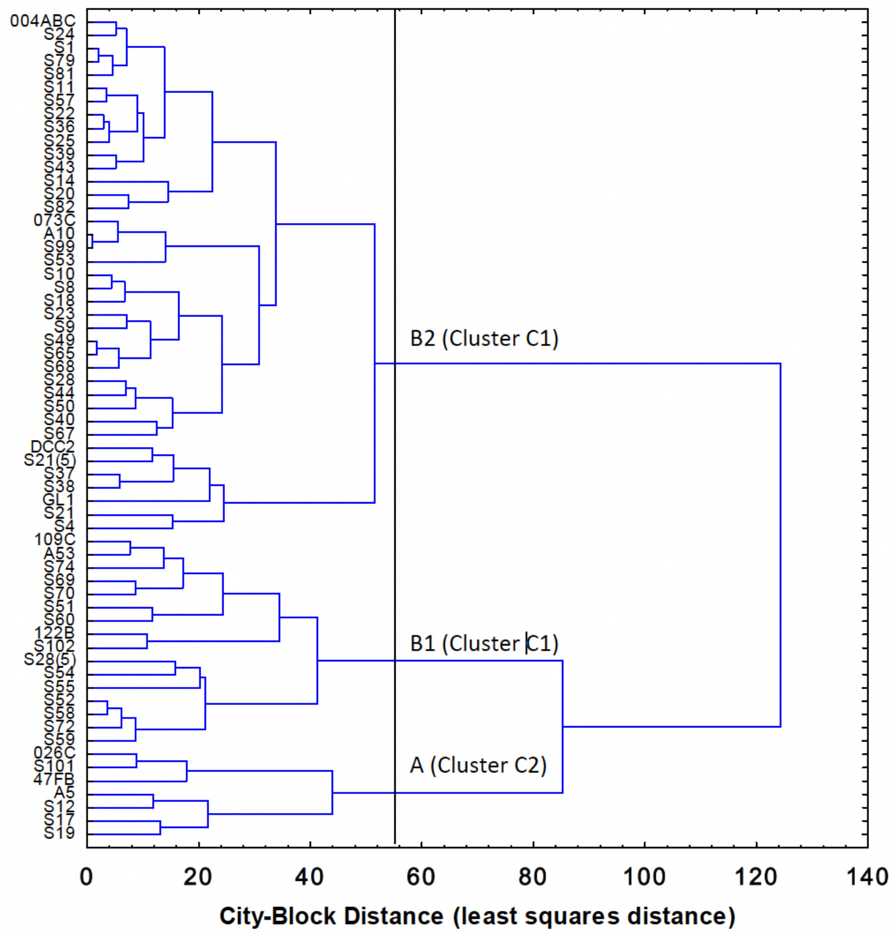


Figure 2 Dendrogram of REF sites (n = 62) grouped according to similar zoobenthic community composition in the 1991, 1999 and 2004/5 Lake Huron-Lake Erie Corridor analysis (Ward's method clustering city-block distances of octave-transformed relative abundances of zoobenthic taxa).

**Figure 1.7:** Branch-selective definition of community types in the 2008 reference dendrogram. The tree first separates $B_2$ from $\{B_1, A\}$, and only at a lower height splits $B_1$ from $A$. Cluster $C_1$ is assigned to $B_2$ and $B_1$ jointly, and Cluster $C_2$ to $A$ alone (red boxes)—a non-monophyletic grouping that no single horizontal cut can produce, since $A$ is topologically closer to $B_1$ than $B_2$ is.

Table 2.5. Summary of observed number of Lake Huron-Lake Erie Corridor sites in each cluster (columns) identified by zoobenthic taxa relative abundances and membership predicted (rows) by discriminant function classification (Appendix III) on the basis of habitat characteristics measured at those sites

| | | Observed | |
|---|---|---|---|
| Group | % Correct | Cluster C1 | Cluster C2 |
| Cluster C1 | 98 | 54 | 1 |
| Cluster C2 | 71 | 2 | 5 |
| Total | 95 | 56 | 6 |

**Figure 1.8:** To do